Sequential Estimation of Network Cascades

Joint work with H. Vincent Poor

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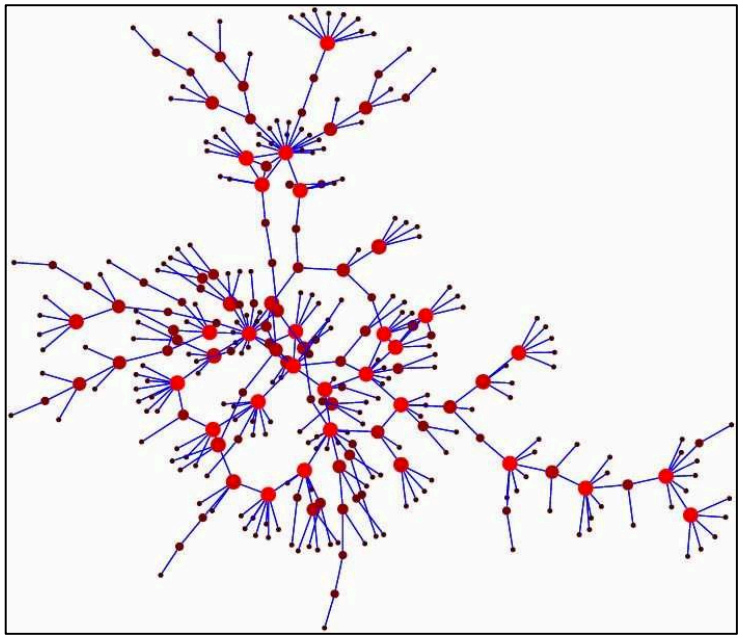
Cascading failure in networks



An epidemic network of HIV transmission[1].

- Anomalies/infections of a small set of individuals spreads rapidly throughout the network
- Observed in diverse settings:
  - Spread of an epidemic in contact networks[2]
  - Viruses in computer networks[3]
  - Fake news in social networks[4]

Q: How quickly can we estimate the cascade source in real time?

1. Ghoshal, Gourab, "Structural and Dynamical Properties of Complex Networks", PhD Dissertation, 2009.
2. N. Antulov-Fantulin, A. Lancic, T. Smuc, H. Stefancic, and M. Sikic, "Identification of patient zero in static and temporal networks: Robustness and limitations," Phys. Rev. Lett., vol. 114, p. 248701, Jun 2015.
3. J. O. Kephart and S. R. White, "Directed-graph epidemiological models of computer viruses," in Proceedings. 1991 IEEE Computer Society Symposium on Research in Security and Privacy, May 1991, pp. 343– 359.
4. H. Allcott and M. Gentzkow, "Social Media and Fake News in the 2016 Election," Journal of Economic Perspectives, vol. 31, no. 2, pp. 211–236, May 2017.

Related Work

Source estimation from a noiseless snapshot

- Cascade evolves stochastically via the SI model from a single vertex and the set of infected nodes is perfectly observed at a future time T.
- Shah and Zaman (2011,2016) compute the maximum likelihood estimator (MLE) for the source and study its properties when the network is a tree
- Estimation for other cascade models – Zhu, Ying (2013-2018)
- Confidence sets for the source – Khim, Loh (2016)

In real-time scenarios:

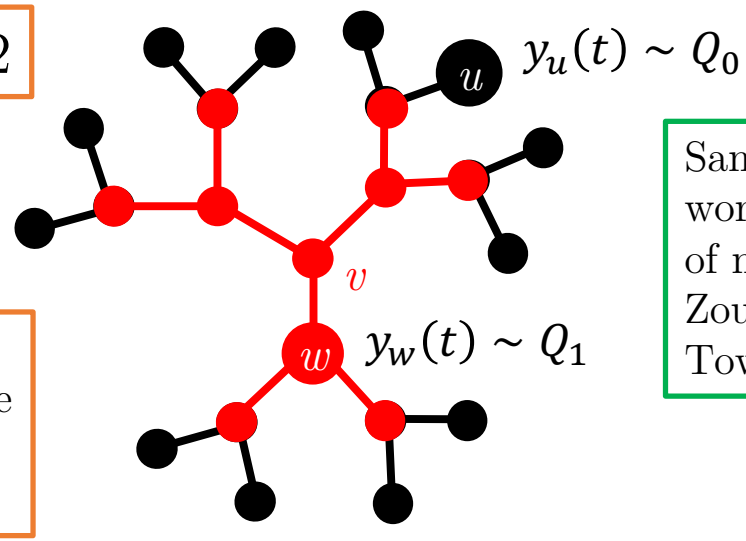
- Infection information often noisy or incomplete, especially in early propagation of the cascade
- Temporal or streaming nature of data can help make better inferences

The Source Estimation Problem

Let $G = (V, E)$ be the graph on which the cascade spreads. We assume the following simple, discrete-time cascade dynamics:

- At time $t = 0$, a single vertex v is infected;
- At time $t > 0$, every vertex within graph distance t of v is infected.

$t = 2$



Same model used in recent work on quickest *detection* of network cascades – see Zou, Veeravalli, Li, Towsley (2019)

cannot observe the true spread, but instead observe signals $\{y_u(t)\}_{t \geq 0}$ for each vertex u

Fundamental tradeoff between estimator accuracy and number of infections.

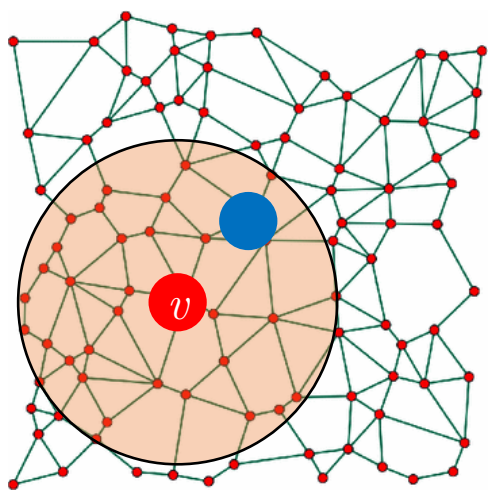
Q: What estimation algorithm achieves the *optimal* tradeoff?

Optimal source estimation

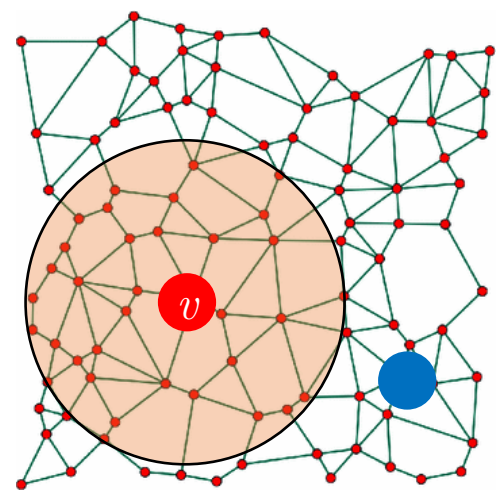
Any sequential estimation procedure can be represented by a **stopping time** T and a **terminal estimate** $D(T)$.

Specify a confidence radius R and Type I error bound α.

Success: $d\big(v, D(T)\big) \leq R$



Failure: $d\big(v, D(T)\big) > R$



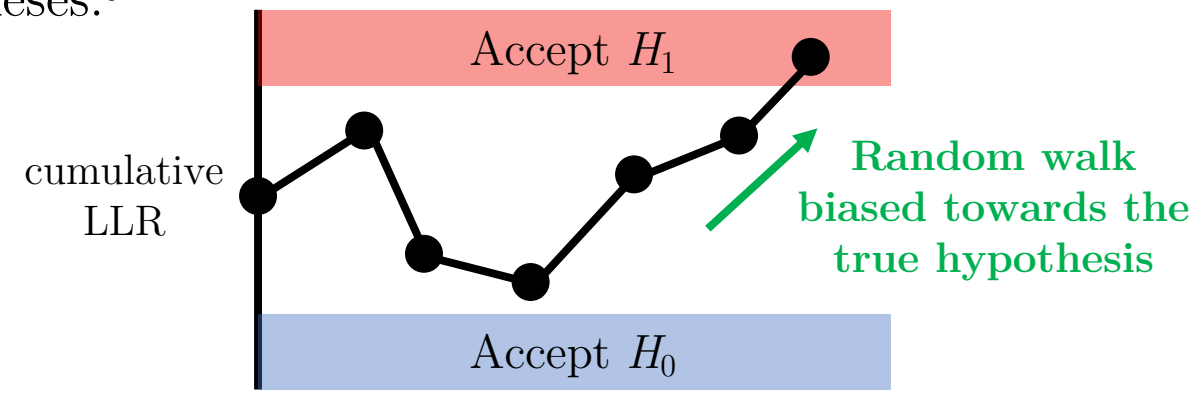
Goal: Solve $T^*(n, R, \alpha) := \min_{(D,T)} \mathbb{E}[T]$ s.t. $\mathbb{P}(failure) \leq \alpha$

A hypothesis testing approach

Key idea: Source estimation is <u>equivalent</u> to the multi-hypothesis testing problem of distinguishing between $\{H_v\}_{v \in V_n}$ where H_v is the hypothesis that v is the cascade source.

Rich prior literature on optimal sequential hypothesis testing:

- The sequential probability ratio test (SPRT) is optimal when there are two hypotheses.[5]



- Appropriate generalizations of the SPRT to multiple (non-homogenous) hypotheses are *asymptotically optimal* as $\alpha \to 0$.[6,7,8]

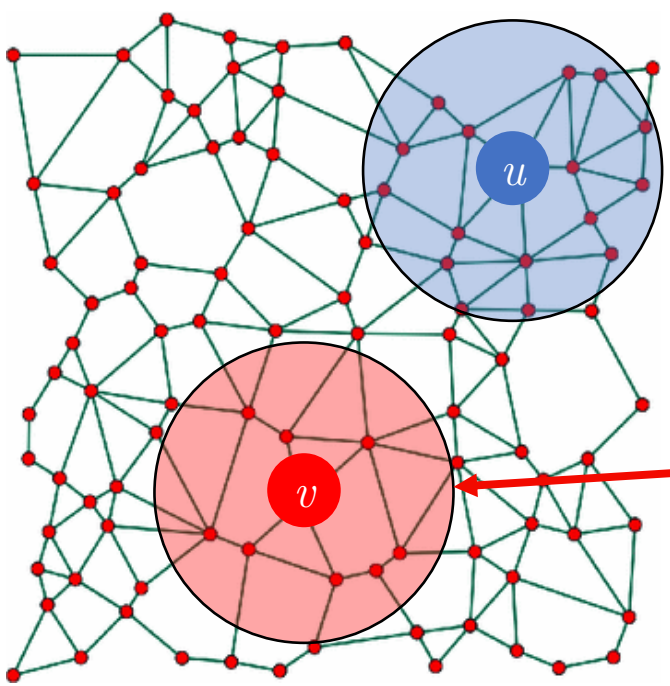
5. A. Wald and J. Wolfowitz, "Optimum character of the sequential probability ratio test," Ann. Math. Statist., vol. 19, no. 3, pp. 326–339, Sept 1948.
6. A. G. Tartakovsky, "Asymptotic optimality of certain multihypothesis sequential tests: Non i.i.d. case," Statistical Inference for Stochastic Processes, vol. 1, no. 3, pp. 265–295, Oct 1998.
7. C. W. Baum and V. V. Veeravalli, "A sequential procedure for multihypothesis testing," IEEE Transactions on Information Theory, vol. 40, no. 6, pp. 1994–2007, Nov 1994.
8. V. V. Veeravalli and C. W. Baum, "Asymptotic efficiency of a sequential multihypothesis test," IEEE Transactions on Information Theory, vol. 41, no. 6, pp. 1994–1997, Nov 1995.

A SPRT-like source estimator

measuring density of anomalies around u

$$Z_u(t) := \sum_{s=0}^{t} \sum_{w \in \mathcal{N}_u(s)} \log \frac{dQ_1}{dQ_0}(y_w(s))$$

cumulative over all data until time t



$Z_u(t)$ small for u
far from the source

$Z_v(t)$ large for v
close to the source

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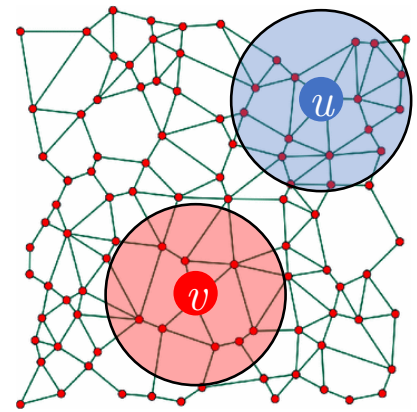
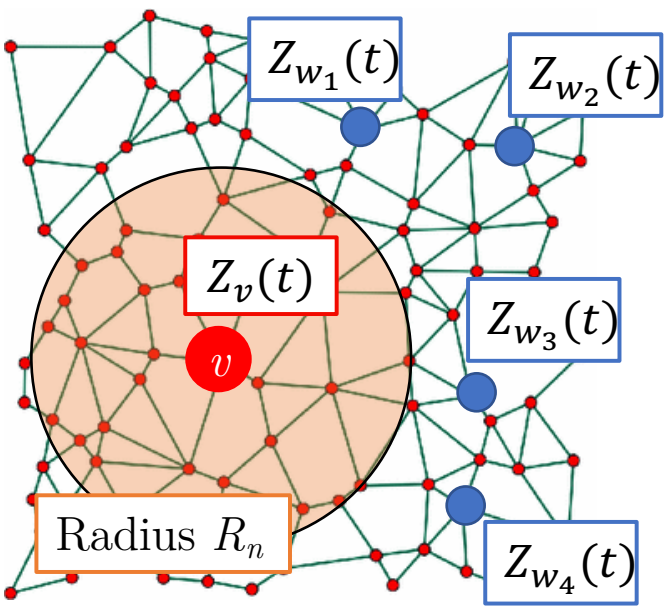
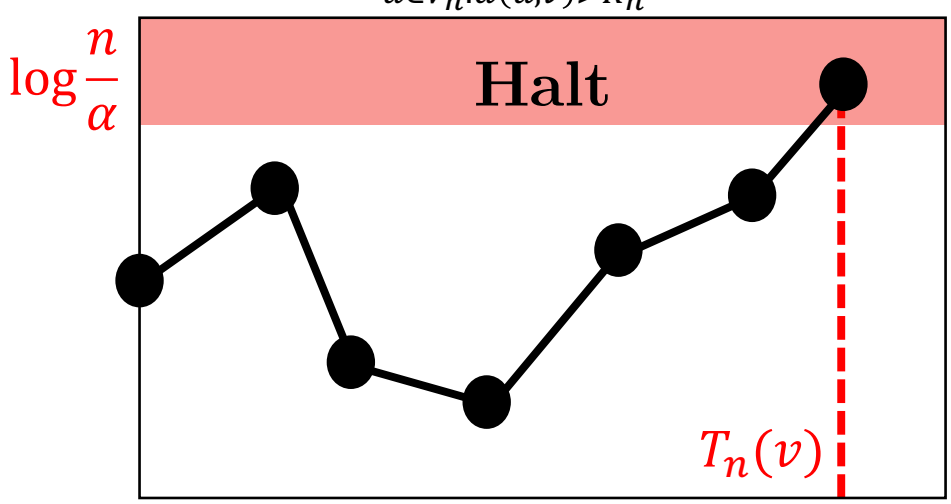


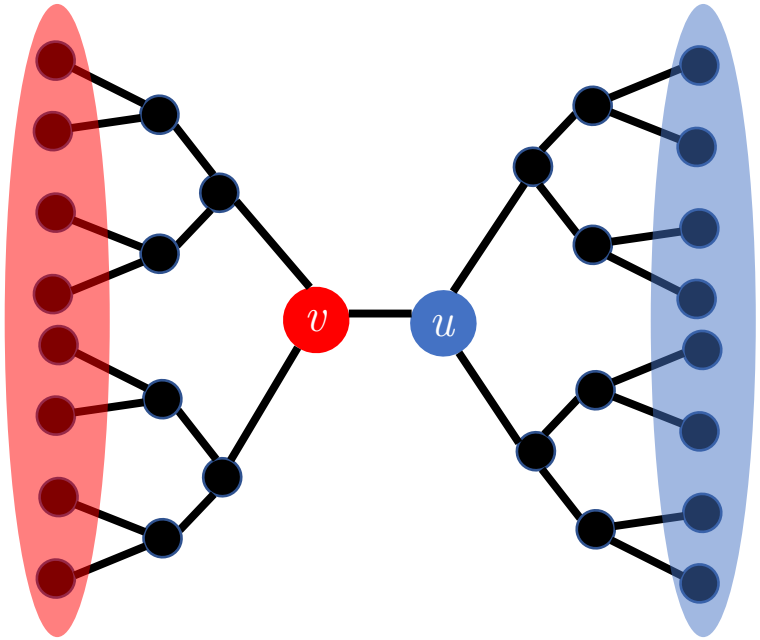


Trajectory of $\min_{u \in V_n : d(u,v) > R_n} (Z_v(t) - Z_u(t))$



Stopping time is $T_n = \min_{v \in V_n} T_n(v)$, source estimate is $D_n = \mathrm{argmin}_{v \in V_n} T_n(v)$

Source estimation in trees



contributions to $Z_v(t) - Z_u(t)$

If G is k-regular and the source is v

- $Z_v(t) - Z_u(t) \asymp (k-1)^t$
- $t \geq \frac{\log \log \frac{n}{\alpha}}{\log(k-1)} \Rightarrow Z_v(t) - Z_u(t) \geq \log \frac{n}{\alpha}$

$$Z_v(t) - Z_u(t) = \log \frac{d\mathbb{P}(\cdot \mid H_v)}{d\mathbb{P}(\cdot \mid H_u)}$$

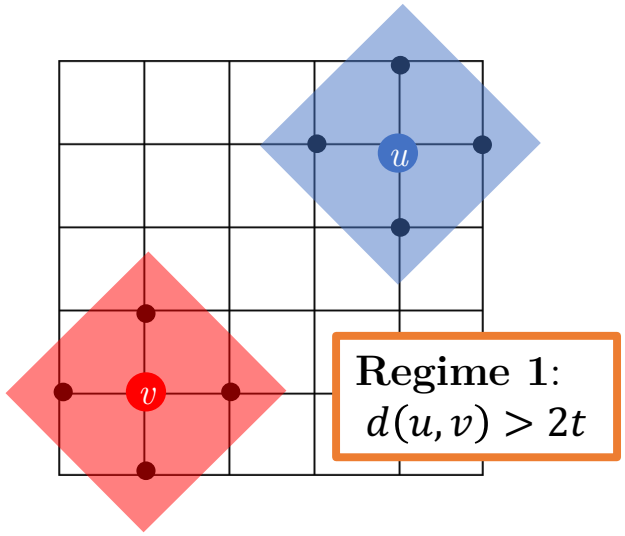
Theorem: Asymptotically optimal estimation in regular trees

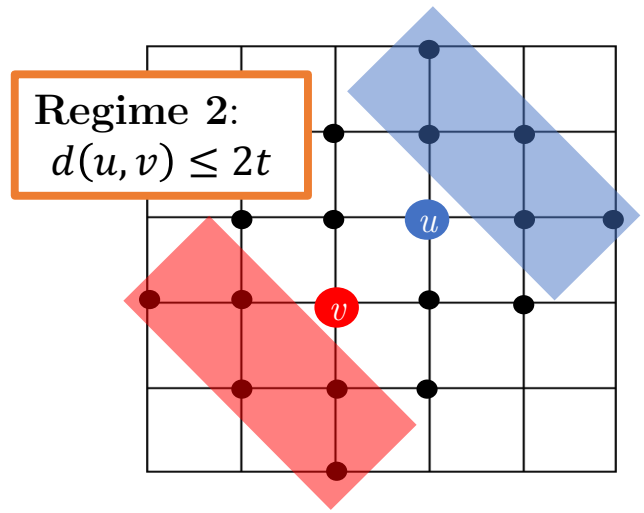
If $R_n \ll \log n$ then (D_n, T_n) is asymptotically optimal as $\alpha \to 0$, and

$$\max_{v \in V_n} \mathbb{E}_v[T_n] \approx_{n,\alpha} T^*(n, R_n, \alpha) \approx_{n,\alpha} \frac{\log \log n}{\log(k-1)}$$

send $n \to \infty$ then $\alpha \to 0$

Source estimation in lattices





$Z_v(t) - Z_u(t) \asymp t^{k+1}$ under H_v

$\Rightarrow$ Need $t \geq \left(\log \frac{n}{\alpha}\right)^{\frac{1}{k+1}}$

$Z_v(t) - Z_u(t) \asymp R_n t^k$ under H_v

$\Rightarrow$ Need $t \geq \left(\frac{1}{R_n} \log \frac{n}{\alpha}\right)^{\frac{1}{k}}$

Theorem: Orderwise optimality in lattices

Let G be the k-dimensional lattice.

If $R_n \gg (\log n)^{\frac{1}{k+1}}$ & $\max_{v \in V_n} |\mathcal{N}_v(R_n)| \ll n$:

$$T^*(n, R_n, \alpha) \asymp_n (\log n)^{\frac{1}{k+1}}$$

If $R_n \ll (\log n)^{\frac{1}{k+1}}$:

$$T^*(n, R_n, \alpha) \asymp_n \left(\frac{\log n}{R_n}\right)^{1/k}$$

Summary

- Want to estimate the source of the cascade *as fast as possible*, given a noisy time-series of the spread
- Derived an orderwise optimal source estimator
- Time to reliable estimation is $\log\log n$ in regular trees, $(\log n)^{\frac{1}{k+1}}$ or $\left(\frac{\log n}{R_n}\right)^{\frac{1}{k}}$ in k-dimensional lattices

Questions?

arXiv link: https://arxiv.org/abs/1912.03800

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